Course Plan: The World of Engineering	
Course No.:	ES 117
Credits: (L-T-P-C)	0-0-3-2
Instructor	**Himanshu Shekhar** Email: himanshu.shekhar@iitgn.ac.in Office: Academic Block 13/321-D
Teaching Assistants	Manu K.S,. Vivek Sharma, Aryan, Gyanendrajit Thockchom, Khomendra Sahu, Abhilipsa Sahu and others

Course contents:

Introduction to Engineering: The past: Case studies of engineering marvels; The present: State of-the-art practices in modern engineering; The future: Identifying and conceptualizing the problem; Model and prototypes: Scale aspects; Design standards; Frugal innovation and logical thinking: Overlaps and tradeoffs; Engineering by Design (Active Mode): C2C: Concept to Commissioning: Ideation and thinking; Engineering project proposal; Budgeting and procurement; From ideas to feasibility (Active Mode): Introduction to simulation engines and programming tools; Feasibility studies; Quantifying milestones; From feasibility to scaled models (Active Mode): Translating ideas and simulations to scaled models; Quantifying performances; Pitching (Active Mode): An elevator pitch to sell/attract investors.

Texts/References: This is a course offered in active mode with a lot of interdisciplinary examples, case studies, and modeling prototyping exercises undertaken to introduce students to the world of engineering. Hence, instead of a prescriptive textbook, a few references are recommended:

a. Moaveni, Saeed. Engineering Fundamentals: An Introduction to Engineering. Cengage Learning, 2019.

b. Xue, Dingyü, and Yang Chen. System Simulation Techniques with MATLAB and Simulink. John Wiley & Sons, 2013.

Learning Outcomes:

1. Get a purview of the world of engineering and how the field of engineering has changed and continues to change to meet societal needs
2. Learn the logical approach to solve the problem in active learning mode
3. Understand and appreciate the complexity and interdisciplinary involved in solving engineering problems
4. Learn to work in teams on time-bound projects with defined objectives
5. Get exposed to the simulation tools available for simulations.
6. Gain hands-on experience in budgeting and acquisition.
7. Learn how to disseminate the information and advance through pitching sessions.

Grading - To be announced. Will include final presentations, weekly summary assignments, and attendance (5%)